

Deep Learning for Ultrasound Thyroid Nodule Classification: A Comparative Study

Abstract— Deep learning has emerged as a promising approach for thyroid nodule classification from ultrasound images. However, many studies primarily report performance on internal test sets derived from the same source as the training data, providing limited evidence of model reliability under independent acquisition conditions. This study analyzes the external generalization capability of six ImageNet architectures, namely MobileNetV2, EfficientNet-B0, ResNet-50, DenseNet-121, ConvNeXt-Tiny, and Swin Transformer-Tiny, for binary thyroid nodule classification. A leakage-safe version of the TN5000 dataset containing 4,874 images was created through exact-duplicate auditing and used for model development, while external validation was performed on the complete TN3K/TNCD cohort of 3,493 images without retraining, threshold adjustment, or recalibration. Operating thresholds were selected on the TN5000 validation set using Youden's J statistic and applied unchanged during both internal and external evaluation. ConvNeXt-Tiny achieved the highest external performance with a ROC-AUC of 0.790, while DenseNet-121 produced comparable results with a smaller generalization gap. All evaluated models showed lower performance on the external cohort, indicating that internal testing alone may overestimate deployable performance. Calibration analysis further revealed that probability reliability remained affected after cross-dataset transfer despite post-hoc calibration. These findings highlight the importance of external validation, leakage auditing, locked-threshold evaluation, and calibration-aware assessment for developing reliable thyroid ultrasound artificial intelligence systems.

Keywords— *Thyroid nodule classification, ultrasound imaging, deep learning, external validation, model calibration, generalization*

I. INTRODUCTION

A high prevalence of thyroid nodules is seen in thyroid sonograms and a correct diagnosis of benign or malignant nodules is crucial for proper management of the disease [1]. The advantages of ultrasound are non-invasive, cost-effective and readily available, hence its wide use. The image quality, acquisition, operator experience, and radiologist interpretation, however, still affect diagnostic performance [2]. While standardized systems and biopsy results aid in diagnosis, there are still difficulties in interpretation when applied to routine clinical practice [2], [3].

Advances in Artificial Intelligence (AI) in recent years have made it possible to develop automated systems for the classification of thyroid nodules [4]. Convolutional Neural Networks (CNNs) and transformer-based image classification models have been demonstrated to have good performance in learning discriminative features directly from ultrasound images [2], [3], [5], [6]. Several studies have shown good classification results by using pre-trained deep learning (DL) architectures and transfer learning strategies. The results show the feasibility of using AI-enabled systems for thyroid cancer screening and diagnosis. But the majority of published studies test their models with an internal test set, which came from the same data as did the training data [5].

One of the important challenges in medical image analysis is to ensure that the DL models perform well in an independent dataset. During modeling, unhelpful characteristics of the dataset that are unique to the imaging device, acquisition method, preprocessing, or annotation may be inadvertently learned, rather than disease-related patterns. This means that models that perform well in their original institution or patient group might have poorer performance when tested in other institutions or patient cohorts [7], [8]. External validation is reported less than architectural development and internal performance evaluation in thyroid ultrasound research. In addition, model calibration and threshold transportability are not frequently discussed, but they are also critical for successful clinical application [7].

Medical AI reporting guidelines have come into focus in recent years and highlight the need for validation in external datasets, calibration, and clear reporting [9], [10]. The receiver operating characteristic area under the curve (ROC-AUC) indicates good differentiation performance but does not guarantee accurate predictions or a performance threshold established at model development time that will work well in a new data distribution. Thus, assessment of discrimination, calibration and operating point performance is needed in order to provide a more complete assessment of model reliability.

In this context, the current study compares six DL architectures in relation to their respective performance: MobileNetV2, EfficientNet-B0, ResNet-50, DenseNet-121, ConvNeXt-Tiny and Swin Transformer-Tiny, under a leakage-safe and externally validated evaluation framework. TN5000 is used for model development, threshold selection, and calibration, while TN3K/TNCD is only used for independent external validation. During internal and external evaluation, the operating thresholds chosen in the validation set are used as is. This experimental design allows direct assessment of cross-dataset generalization, threshold transportability and calibration without external retraining, optimization or recalibration.

The contributions of this study are summarized as follows:

- Classification is performed using exact duplicate detection and removal to build a leakage-safe thyroid ultrasound classification framework.
- Six of the most popular DL architectures are systematically evaluated for benign and malignant thyroid nodule classification.
- External validation is done by an independent dataset to test cross-dataset generalizability.
- An evaluation approach using a locked threshold is used to study transferability of thresholds across various data distributions.
- The performance of the models is thoroughly assessed by applying discrimination, calibration, computational efficiency and error analysis.

The remainder of this study is organized as follows. Section II reviews related work on thyroid ultrasound classification. Section III describes the datasets, DL architectures, training approach, evaluation metrics and experimental procedures. Section IV presents the experimental results. Finally, Section V discusses the findings and concludes the study.

II. RELATED WORK

DL is massively used for the classification of thyroid nodules in ultrasound images. DL architectures like ResNet, VGG, Inception, EfficientNet, DenseNet and GoogLeNet have shown promising classification performance and have often performed better than the conventional radiomics based methods [5], [11]. The results of these studies illustrate that DL models can be developed and trained successfully to learn discriminative ultrasound features for the classification of benign and malignant thyroid nodules.

In addition to the basic CNNs architectures, recent research has explored attention mechanisms, ensemble learning, multimodal ultrasound analysis, handcrafted radiomic features, the use of clinical information, and explainable artificial intelligence approaches like Grad-CAM [12], [13]. Most of these methods have claimed high classification rates and illustrate the promise of AI as a decision support tool for thyroid ultrasound diagnosis [12], [14], [15].

In spite of these developments, most studies focus on in-house evaluations of the models. The results of models tested using data from the same source as the models were developed from may not reflect the actual performance of a model in clinical settings. Thus, the external validations of the model's reliability and generalizability are regarded as an important step [7]. The relevance of the model results for external use may be reduced as demonstrated by previous imaging studies, which typically have experienced reduced model performance when the model was tested with new data from other imaging protocols, different patient populations, or different types of datasets [8], [16]. This has been observed also in the classification of thyroid ultrasound images, where the models might have lower performance on independent datasets [17].

In addition to the external validation, model calibration and threshold identification are also crucial for clinical use. A well discriminating model can be used to give poorly reliable probabilities or inappropriate sensitivity and specificity in a new distribution of data. These aspects are however, assessed less regularly than the classification accuracy and architectural performance.

The current research works are mostly about new architectures or about the internal performance enhancement. This study, however, focuses on the reliability of the evaluation, using an experimental design that prevents leakage, testing on an independent cohort, a locked threshold testing, and a calibration analysis [18]. This assessment offers a more thorough measure of model performance in realistic cross-dataset settings.

III. METHODOLOGY

This study presents a leakage-safe external validation framework for thyroid nodule classification using thyroid ultrasound images. The TN5000 dataset is used for model training, threshold selection, calibration analysis, and internal evaluation, whereas the TN3K/TNCD dataset is reserved

exclusively for external validation. The overall experimental workflow is illustrated in Fig. 1.

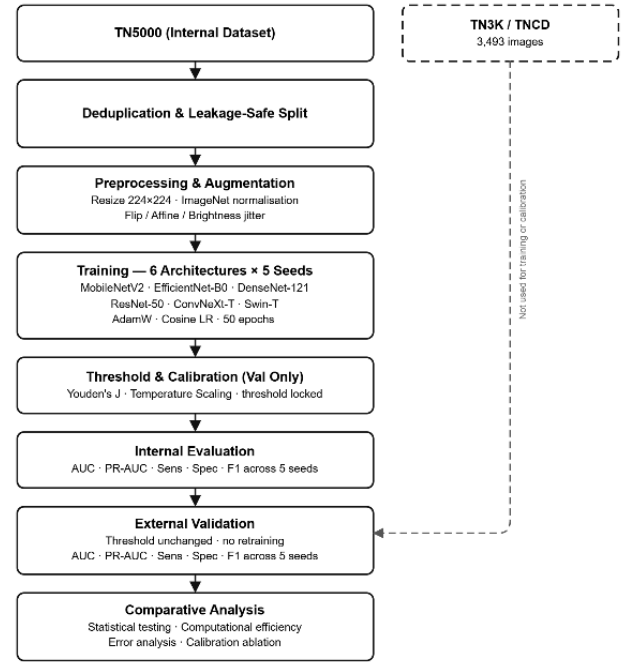


Fig. 1 Illustration of study workflow

A. Datasets

This study uses two publicly available thyroid ultrasound datasets. The TN5000 dataset contains 5,000 B-mode thyroid ultrasound images with expert annotations and biopsy-confirmed benign or malignant labels [19]. The original dataset includes 3,500 training images, 500 validation images, and 1,000 test images. To ensure a leakage-safe evaluation, an MD5 hash audit was performed across the entire dataset. This process identified 119 duplicate-image groups containing 245 images, including 65 groups distributed across multiple dataset partitions. After removing all duplicate images, 4,874 images remained for model development and internal evaluation.

External validation was performed using the TN3K/TNCD dataset [20], [21], which contains thyroid ultrasound images, segmentation masks, and binary classification labels. Only the ultrasound images were used as model inputs, while the segmentation masks were excluded from model training and classification. The complete external cohort consisted of 3,493 images, including 2,283 benign and 1,210 malignant nodules. The original training, validation, and test partitions were combined and reserved exclusively for external validation. A summary of both datasets is presented in Table I.

TABLE I. DATASET OVERVIEW

Dataset	Purpose	N	Benign	Malignant
TN5000 Train	Model Training	3,448	1,018	2,430
TN5000 Validation	Threshold Selection & Calibration	476	116	360
TN5000 Locked Test	Internal Evaluation	950	257	693
TN3K/TNCD	External Validation	3,493	2,283	1,210

B. Deep Learning Architectures

Six DL architectures were selected to represent different model families commonly used for medical image classification. These include MobileNetV2, EfficientNet-B0, ResNet-50, DenseNet-121, ConvNeXt-Tiny, and Swin Transformer-Tiny. MobileNetV2 and EfficientNet-B0 are lightweight architectures designed for computational efficiency. ResNet-50 and DenseNet-121 are widely adopted CNN models with strong feature extraction capability [6]. ConvNeXt-Tiny represents a modern convolution-based architecture, whereas Swin Transformer-Tiny is a vision transformer that learns both local and global image representations.

The objective of this study is to systematically compare these representative DL architectures using a unified experimental framework. Their performance is evaluated in terms of classification accuracy, generalization capability, calibration quality, and computational efficiency to provide a comprehensive assessment of their suitability for thyroid ultrasound nodule classification.

C. Training Strategy

All models were trained on the TN5000 training set using transfer learning. To improve training stability and reduce the influence of random initialization, each architecture was trained independently using five random seeds. Model selection was based exclusively on performance on the TN5000 validation set. The external TN3K/TNCD cohort was not used during model training, hyperparameter tuning, threshold selection, or calibration fitting.

After training, each model was evaluated on the locked TN5000 test set to measure internal performance. The trained models were then directly evaluated on the complete TN3K/TNCD dataset without retraining, adaptation, or parameter updates. This evaluation strategy provides an unbiased assessment of model generalization on an independent external dataset.

D. Evaluation Metrics

Model performance was evaluated using Accuracy, Sensitivity, Specificity, Precision, and F1-score. These metrics were used to assess classification performance and the ability of the models to distinguish between benign and malignant thyroid nodules. ROC-AUC and Precision-Recall Area Under the Curve (PR-AUC) were also computed to evaluate discrimination performance. The evaluation metrics are defined in (1) to (5) as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Sensitivity = \frac{TP}{TP + FN} \quad (2)$$

$$Specificity = \frac{TN}{TN + FP} \quad (3)$$

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

$$F1 = \frac{2 (Precision \times Sensitivity)}{Precision + Sensitivity} \quad (5)$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively. In this binary classification task, Sensitivity is equivalent to Recall. Higher values of Accuracy, Sensitivity, Specificity, Precision, F1-score, ROC-AUC, and PR-AUC indicate better classification and discrimination performance.

E. Threshold Selection

The operating threshold for each model was determined using Youden's J statistic on the TN5000 validation set. This approach identifies the threshold that provides the best balance between sensitivity and specificity. The selected threshold was fixed and applied without modification to both the locked TN5000 test set and the TN3K/TNCD external cohort. No threshold optimization was performed using the external labels. Youden's J statistic is defined in (6) as follows:

$$J = Sensitivity + Specificity - 1 \quad (6)$$

This evaluation strategy enables an unbiased assessment of threshold transportability across an independent external dataset.

F. Calibration Analysis

Probability calibration was evaluated using the Brier Score and Expected Calibration Error (ECE). The Brier Score measures the difference between predicted probabilities and observed outcomes, while ECE quantifies the agreement between model confidence and prediction accuracy. These metrics are defined in (7) and (8) as follows:

$$Brier = \left(\frac{1}{N}\right) \sum_{i=1}^N (p_i - y_i)^2 \quad (7)$$

$$ECE = \sum_{m=1}^M \left(\frac{|B_m|}{N}\right) |acc(B_m) - conf(B_m)| \quad (8)$$

where (N) is the total number of samples (p_i) is the predicted probability for sample (i) , and (y_i) is the corresponding true label. In the ECE calculation (B_m) denotes the (m^{th}) probability bin, $(acc(B_m))$ is the classification accuracy within the bin, and $(conf(B_m))$ is the average prediction confidence. Lower Brier Score and ECE values indicate better probability calibration. Temperature scaling was performed using the TN5000 validation set to improve probability calibration while preserving the classification performance of the models.

G. Statistical and Computational Analysis

Statistical analysis was conducted for measuring model performance and reproducibility. The statistical analysis included the estimation of bootstrap 95% confidence interval

for external performance measures and comparison of models via bootstrap, Monte Carlo permutation test and McNemar exact test with Holm correction. Reproducibility was measured via paired Wilcoxon signed rank test on five random seeds.

Computational efficiency was evaluated using parameter count, checkpoint size, giga floating-point operations per second (GFLOPS), inference latency, and throughput on a Tesla T4 graphics processing unit (GPU).

IV. RESULTS

The performance of the six evaluated DL models was assessed using the TN5000 locked test set for internal evaluation and the TN3K/TNCD cohort for external validation. Results are reported as median values across five independent random seeds. All models showed lower performance during external validation, indicating reduced generalization after transfer to an independent dataset.

ConvNeXt-Tiny achieved the highest external ROC-AUC of 0.790 and PR-AUC of 0.700, followed closely by DenseNet-121 with an external ROC-AUC of 0.787 and PR-AUC of 0.696. DenseNet-121 exhibited the smallest ROC-AUC gap between internal and external evaluation (0.137), while Swin Transformer-Tiny showed the largest reduction (0.179). The internal and external discrimination performance of all evaluated models is summarized in the Table II.

DenseNet-121 achieved the highest external accuracy (0.757) and specificity (0.832), while ConvNeXt-Tiny achieved the highest sensitivity (0.652) and F1-score (0.645). Overall, ConvNeXt-Tiny and DenseNet-121 consistently outperformed the remaining architectures across both discrimination and operating-point metrics. The external operating-point performance is summarized in Table III.

TABLE II. INTERNAL AND EXTERNAL MODEL PERFORMANCE

Model	Internal ROC-AUC	External ROC-AUC	ROC-AUC Gap	PR-AUC
ConvNeXt-Tiny	0.941	0.790	0.151	0.700
DenseNet-121	0.924	0.787	0.137	0.696
ResNet-50	0.915	0.751	0.164	0.651
EfficientNet-B0	0.899	0.751	0.148	0.642
Swin Transformer-Tiny	0.911	0.732	0.179	0.606
MobileNetV2	0.870	0.725	0.145	0.595

TABLE III. EXTERNAL CLASSIFICATION PERFORMANCE

Model	Accuracy	Sensitivity	Specificity	F1-Score
ConvNeXt-Tiny	0.756	0.652	0.818	0.645
DenseNet-121	0.757	0.637	0.832	0.643
ResNet-50	0.729	0.574	0.822	0.614
EfficientNet-B0	0.718	0.579	0.795	0.595
Swin Transformer-Tiny	0.711	0.638	0.758	0.575
MobileNetV2	0.690	0.534	0.774	0.545

A. External ROC and Precision-Recall Analysis

The discrimination performance of the evaluated models was further assessed using Receiver Operating Characteristic

(ROC) and Precision-Recall (PR) curves on the TN3K/TNCD cohort. ConvNeXt-Tiny and DenseNet-121 demonstrated the strongest discrimination capability, while ResNet-50 and EfficientNet-B0 showed intermediate performance. Swin Transformer-Tiny and MobileNetV2 achieved lower discrimination performance. Similar trends were observed in the PR analysis, indicating stronger malignant nodule identification by the leading models. The external ROC and PR curves are shown in Fig. 2 and Fig. 3, respectively.

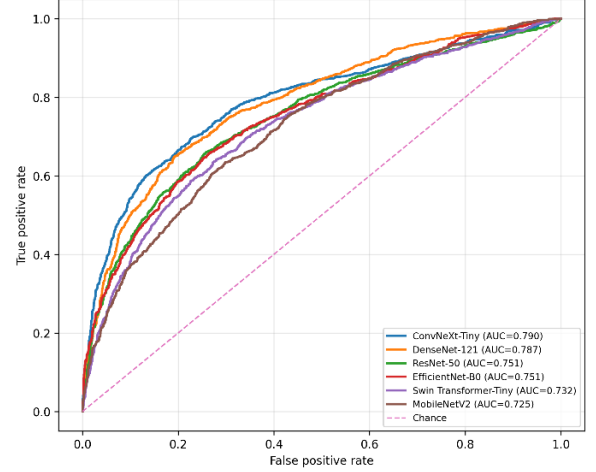


Fig. 2 External ROC curves for all evaluated models on the complete TN3K/TNCD cohort

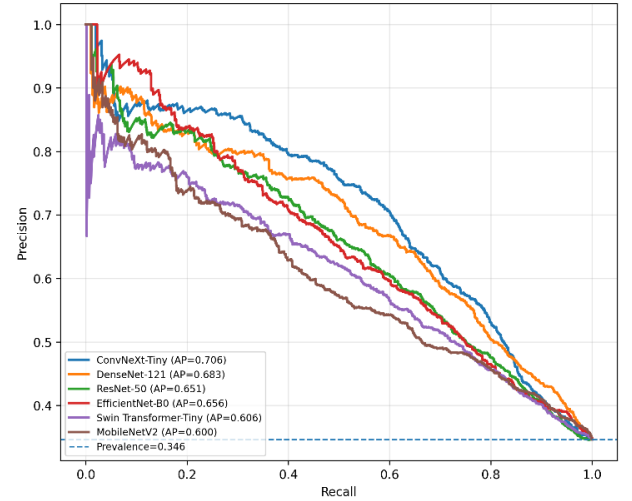


Fig. 3 External precision-recall curves on TN3K/TNCD

B. Internal-to-External Generalization Gap

The model robustness was assessed by comparing model internal and external ROC-AUC to all the architectures. For all models a performance drop was noticed in the external validation showing the effect of the dataset shift on the classification performance. Some models had a greater reduction in their internal performance level at the independent cohort while others did not. The reduction was the lowest for DenseNet-121 and the highest for Swin Transformer-Tiny. The results underscore the need for external validation to evaluate model reliability and applicability. In comparison, the internal and external ROC-AUC values are presented in Fig. 4.

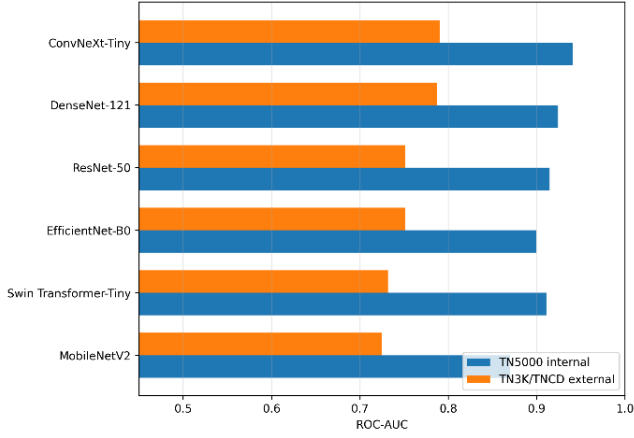


Fig. 4 Internal vs external ROC-AUC comparison across all evaluated architectures

C. Confusion Matrix Analysis

The analysis of the confusion matrix was used to analyze the classification behavior at the selected operating threshold. The representative ConvNeXt-Tiny model identified 1,868 benign nodules and 774 malignant nodules correctly and 415 false positives and 436 false negatives were seen. Important is the fact that there are also false negative results, which can decrease the efficiency of screening and early diagnostic assistance. These results reveal that the thresholds derived from the model development might not be directly applicable to other data sets and should be tested and validated carefully prior to use. The results of the Confusion matrix of ConvNeXt-Tiny are displayed in Fig. 5.

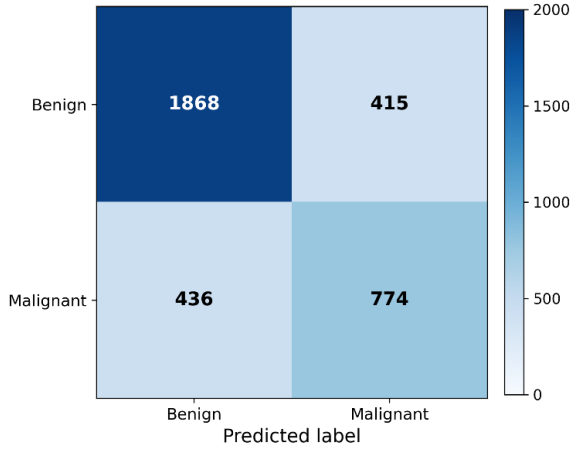


Fig. 5 TN3K/TNCD confusion matrix for ConvNeXt-Tiny

D. Calibration Analysis

Calibration curves, Brier Score and Expected Calibration Error (ECE) were used to assess calibration performance. The reduction in the difference between the level of probability reliability predicted and realized has been accomplished through temperature scaling. After calibration, the Brier Score for ConvNeXt-Tiny was reduced to 0.190 while that of ECE was reduced to 0.136. Calibration helped to improve probability estimates but there were deviations from perfect calibration after external transfer, especially at higher probability levels. The results show that there is still a need to exercise caution in the interpretation of calibrated probabilities when using models on independent data sets.

Before and after temperature scaling calibration curves are displayed in Fig. 6.

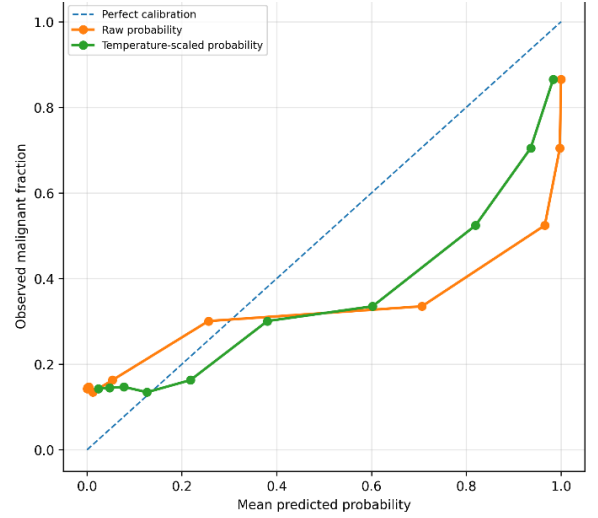


Fig. 6 Calibration curve for ConvNeXt-Tiny on TN3K/TNCD before and after temperature scaling

V. DISCUSSION AND CONCLUSION

This study evaluated six DL models for thyroid nodule classification using a leakage-safe development dataset and an independent external validation cohort. All models showed reduced performance during external validation, indicating that strong internal results do not always translate to independent datasets. ConvNeXt-Tiny achieved the best external discrimination performance, while DenseNet-121 showed the smallest performance reduction between internal and external evaluation. These findings highlight the importance of evaluating both predictive performance and model generalization.

The results also show that discrimination metrics alone are insufficient for clinical deployment. The locked-threshold evaluation indicated that thresholds selected during model development may not remain optimal across different data distributions. Calibration analysis further showed that probability reliability and discrimination performance measure different aspects of model behavior. Although temperature scaling improved calibration quality, residual calibration error remained after external validation.

The analysis was performed at the image level because patient identifiers are limited. In addition, only ultrasound images were used without clinical information, laboratory findings, or radiologist-reported features. Overall, the findings emphasize the importance of external validation, duplicate-leakage auditing, threshold evaluation, and calibration assessment for reliable thyroid ultrasound AI systems. Future work should include patient-level evaluation, prospective multicenter validation, integration of clinical information, comparison with radiologists, and clinically informed threshold optimization.

ETHICAL STATEMENT

This study uses publicly available datasets and no direct human participation or identifiable personal information was involved therefore ethical approval was not required.

DATA AVAILABILITY

Datasets used in this study are publicly available and the code and supporting findings of this study is available from corresponding author upon reasonable request.

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